1 Absolute Convergence

Given any series $\sum_{n=1}^{\infty} a_n$, we can consider the corresponding series

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \cdots$$

whose terms are the absolute values of the original series.

Definition 1.1 (Absolute convergence)**.** A series $\sum_{n=1}^{\infty} a_n$ is said to *converge absolutely* if the series of absolute values $\sum_{n=1}^{\infty} |a_n|$ converges.

Remark 1.2. If $\sum_{n=1}^{\infty} a_n$ is a series whose terms are all positive, then $|a_n| = a_n$, so absolute convergence is the same as convergence in this case.

1. Show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ converges absolutely.

2. Is the alternating harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ absolutely convergent?

Definition 1.3 (Conditional convergence)**.** A series $\sum_{n=1}^{\infty} a_n$ is said to *converge conditionally* if it converges, but does not converge absolutely. [1]

Example 1.4. By the result of question 2, the alternating harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges conditionally.

The previous example shows that it is possible for a series to converge, but not converge absolutely. However, the next theorem shows that absolute convergence implies convergence.

Theorem 1.5 (Absolute convergence implies convergence)**.** *If a series $\sum_{n=1}^{\infty} a_n$ converges absolutely, then it converges.*

Proof. Since $|a_n|$ is either a_n or $-a_n$, we have the inequality

$$0 \leq a_n + |a_n| \leq 2|a_n|.$$

3. Use the inequality above to complete the proof.

□

4. Determine whether the series $\displaystyle\sum_{n=1}^{\infty} \frac{\cos(n)}{n^2}$ converges or diverges.

[1] The question of whether a given convergent series is absolutely convergent or conditionally convergent has a bearing on the question of whether infinite sums behave like finite sums. If we rearrange the order of the terms in a finite sum, then of course the value of the sum remains unchanged. This is not always the case for an infinite series. However, it can be proved that if a series converges absolutely, then any rearrangement of the series leaves the value of the sum unchanged.

2 Ratio Test

The following test is very useful in determining whether a given series is absolutely convergent.

Theorem 2.1 (Ratio Test)**.**

(i) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *is absolutely convergent (and therefore convergent).*

(ii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$ *or* $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty$ *, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*

(iii) If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$*, the Ratio Test is inconclusive; that is, no conclusion can be drawn about the convergence or divergence of the series* $\sum_{n=1}^{\infty} a_n$*.*

Proof. (i) The idea is to compare the given series with a convergent geometric series. Since $L < 1$, we can choose a number r such that $L < r < 1$.

5. Show that $\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right| = L$ implies there exists $N \in \mathbb{N}$ such that for $n \geq N$ we have $\left|\frac{a_{n+1}}{a_n}\right| < r$, or, equivalently, $|a_{n+1}| < |a_n|r$.

6. Show that the result of question 5 implies that $|a_{N+k}| < |a_N|r^k$ for all $k \geq 1$.
(*Hint*: $|a_{N+k}| = \dfrac{|a_{N+k}|}{|a_{N+k-1}|}\dfrac{|a_{N+k-1}|}{|a_{N+k-2}|}\cdots\dfrac{|a_{N+1}|}{|a_N|}|a_N|$)

7. Use the inequality $0 \leq |a_{N+k}| < |a_N| r^k$, $k \geq 1$, to prove that the series $\sum_{k=1}^{\infty} |a_{N+k}|$ converges. Since $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{N} |a_n| + \sum_{k=1}^{\infty} |a_{N+k}|$, this implies the series $\sum_{n=1}^{\infty} |a_n|$ converges, completing the proof of part (i).

(ii) Now suppose $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \infty$. By an argument similar to the one in the proof of question 5, there exists $N \in \mathbb{N}$ such that $n \geq N$ implies $\left|\frac{a_{n+1}}{a_n}\right| > 1$, or, equivalently, $|a_{n+1}| > |a_n|$. Since the terms of the sequence are increasing for all $n \geq N$, $\lim_{n\to\infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} a_n$ diverges by the Divergence Test.
(iii)

8. Show that if $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = 1$, then Ratio Test is inconclusive. (*Hint*: Consider the series $\sum_{n=1}^{\infty} \frac{1}{n}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$.)

□

9. Test the series $\sum_{n=1}^{\infty}(-1)^n\frac{n^3}{3^n}$ for absolute convergence.

10. Test the series $\sum_{n=1}^{\infty}\frac{n^n}{n!}$ for absolute convergence.

3 Root Test

The following test is convenient to apply when nth powers occur.

Theorem 3.1 (Root Test)**.** *(i) If* $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$*, then the series* $\sum_{n=1}^{\infty} a_n$ *is absolutely convergent (and therefore convergent).*

(ii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$ *or* $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \infty$*, then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*

(iii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$*, the Root Test is inconclusive.*

Proof. The proof of the Root Test is similar to the proof of the Ratio Test.

11. (Optional) Prove the Root Test. (*Hint for part (i)*: Choose any number r such that $L < r < 1$ and use the fact that there is an integer N such that $n \geq N$ implies $\sqrt[n]{|a_n|} < r$ whenever $n \geq N$.

□

12. Test the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$.